

Quiz — Section 6.6

1. A cannonball is fired at an angle θ to the horizontal with initial velocity v_0 feet per second. If air resistance is ignored, the horizontal range is

$$R(\theta) = \frac{1}{16}v_0^2 \sin \theta \cos \theta.$$

1. Rewrite $R(\theta)$ in terms of $\sin(2\theta)$.
2. Find the angle θ that maximizes the range.

Hint: $\sin(2\theta) = 2 \sin \theta \cos \theta$

$$1. \quad R(\theta) = \frac{1}{16}v_0^2 \sin \theta \cos \theta = \frac{1}{16}v_0^2 \cdot \frac{1}{2} \sin(2\theta) = \frac{1}{32}v_0^2 \sin(2\theta)$$

2. Range is maximized when $\sin(2\theta) = 1$

$$\text{so } 2\theta = \frac{\pi}{2},$$

$$\text{so } \theta = \frac{\pi}{4}.$$

2. If

$$\sin \theta = \frac{3}{5}, \quad 0 < \theta < \frac{\pi}{2},$$

find the exact value of

$$\cos(2\theta).$$

Hint: $\cos(2\theta) = 1 - 2\sin^2 \theta$

$$\begin{aligned}\cos(2\theta) &= 1 - 2\sin^2 \theta = 1 - 2\left(\frac{3}{5}\right)^2 \\ &= 1 - \frac{18}{25} = \frac{7}{25}.\end{aligned}$$